Developing RSE skills at the novice level

Student RSE model for research support, experiential learning, and technical workforce development

Paula Lackie, Oliver Hall, Tim Dennis, Leigh Phan
Carleton College DataSquad | UCLA Library Data Science Center DataSquad

Paid students contribute to research and build RSE skills while they do it.



Campuses need more data and software help than staff can provide. Courses leave gaps. So does the code researchers write. DataSquads fill those gaps. Many students never hear of this work, especially students from underrepresented groups.

A paid job that teaches

Students are paid employees. Library or department staff supervise them. Students help researchers with their data, from cleaning to reproducibility.

Who is on a DataSquad

Project manager	triages requests
Data analysts and developers	do the hands-on work with data and code
Technical writers	document projects
Staff	supervise, mentor, fund, and sustain the DataSquad

Limits we keep

We help with a researcher's data and code. We don't change their research questions, tutor their students, or share anything a researcher tells us in confidence.

What slows us down

Student schedules, the academic calendar, onboarding, review, and graduation handoffs all slow the work. More students can take more varied projects, but they need more oversight. Handbooks, templates, and staff stay when students graduate.

What the work looks like

CARLETON **Election-news scraper.** Nine students started, extended, or used it from 2021 to 2024 to collect broadcast transcripts for a political science project. **The scraper outlasted the students who began it.**

UCLA **wbcheck.** Students built the first version of this Carpentries lesson checker in 2023. Staff expanded it in 2026 with AI assistance. It has 48 documented rules, tests, a BSD-3 license, and a citation file that credits its student authors (doi:10.5281/zenodo.23329531).

CARLETON **Language placement dashboard.** The project was longer than any student would stay. The DataSquad sketched the database and chose heatmaps for the development team.

UCLA **UC OSPO Education site.** Students built the lesson pages and search for a catalog of 55 open source lessons about open science and research software. The site has a citation file and a DOI (doi:10.5281/zenodo.23330022) that credit its developers.

Research software habits, when the project needs them. Run instructions, a dependency list, one test, and a citation file. We do not treat them as a checklist. We plan to track whether researchers keep them.

Where students go next

We located 19 former UCLA student employees. Their current job categories, from public profiles in 2026:

Data / analytics / consulting	11
Software / engineering / IT	4
Research / graduate school	2
Nonprofit / public sector	2

Descriptive, not a placement rate.

What adopting takes

- Paid student positions and a wage budget
- Supervisors and mentors with time to review and coach
- Onboarding and a home for documentation
- A plan for turnover

Start a DataSquad

Get the starter kit for roles, onboarding, confidentiality, and request tracking. Students learn software sharing from our open lesson, **Sharing Research Software**. Supervise students? Compare onboarding, review, and graduation handoffs with us.

datasquad.info/join